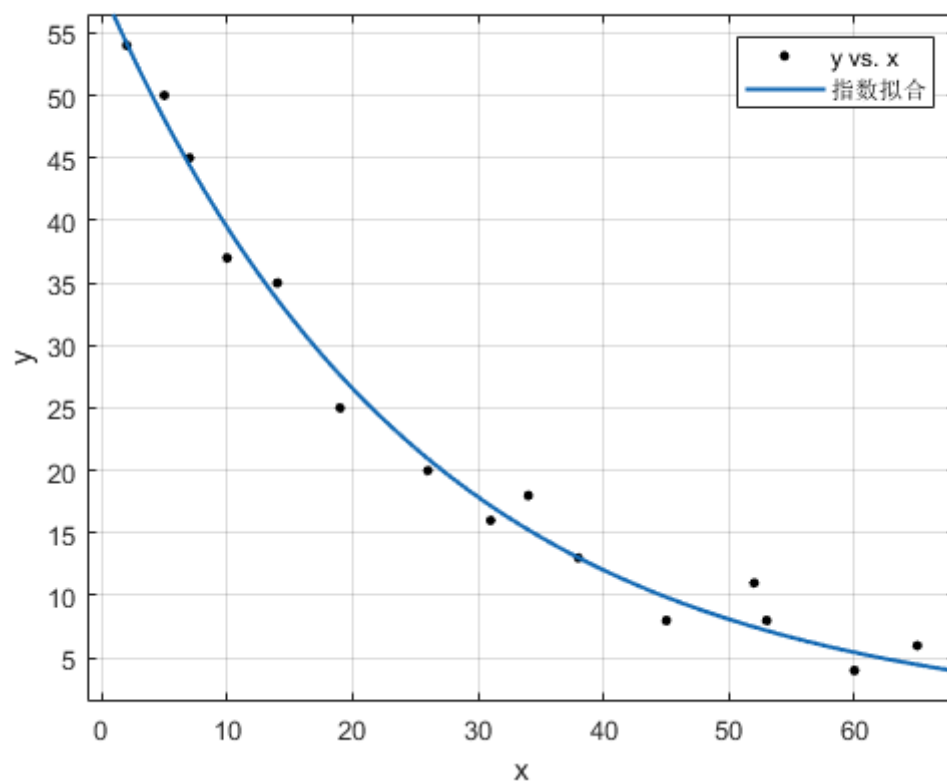
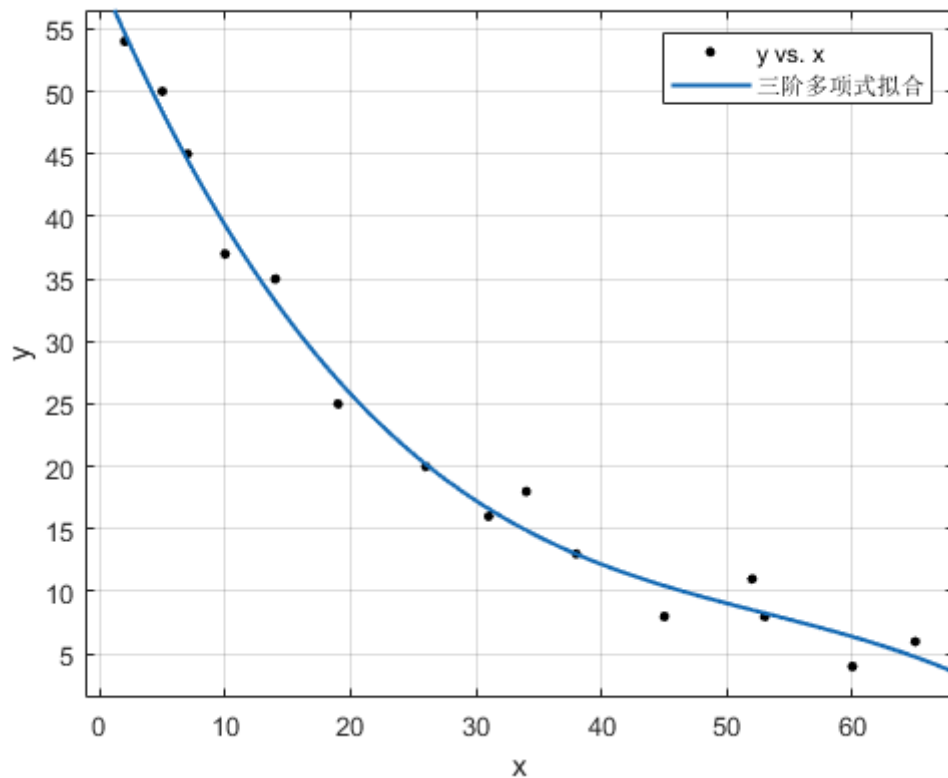


# Numerical Analysis Exercise 9

1.



$R^2 = 0.98746$  ,  $SSE = 49.459$ .



$R^2 = 0.9886$ ,  $SSE = 44.954$ .

2.

```

1  clc
2  clear all
3  close all
4  %%Data insert
5  y=
   [11.2,8.8,12.3,11.6,13.4,18.3,11.1,12.1,9.6,8.4,9.3,10.6,8.4,9.6,10.9,10.1,14.8,9.1,10.8,10.
    2,13.6,14.9];
6  x1=
   [5.68,3.79,6.02,4.85,4.60,6.05,4.90,7.08,3.85,4.65,4.59,4.29,7.97,6.19,6.13,5.71,6.40,6.06,5
    .09,6.13,5.78,5.43];
7  x2=
   [1.90,1.64,3.56,1.07,2.32,0.64,8.50,3.00,2.11,0.63,1.97,1.97,1.93,1.18,2.06,1.78,2.40,3.67,1
    .03,1.71,3.36,1.13];
8  x3=
   [4.53,7.32,6.95,5.88,4.05,1.42,12.60,6.75,16.28,6.59,3.61,6.61,7.57,1.42,10.35,8.53,4.53,12.
    79,2.53,5.28,2.96,4.31];
9  x4=
   [8.2,6.9,10.8,8.3,7.5,13.6,8.5,11.5,7.9,7.1,8.7,7.8,9.9,6.9,10.5,8.0,10.3,7.1,8.9,9.9,8.0,11
    .3];
10
11
12  X=[ones(size(y,2),1)';x1;x2;x3;x4];
13  X = X';
14  y= y';
15  [b,bint,r,rint,s]=regress(y,X)
16

```

```

1  b = 5×1
2      7.0294
3      -0.7598
4      0.3670
5      -0.2860
6      1.0589
7
8  bint = 5×2
9      1.9343    12.1246
10     -1.6159    0.0963
11     -0.1549    0.8888
12     -0.5171   -0.0550
13      0.5615    1.5564
14
15  r = 22×1
16      0.4015
17     -1.1646
18     -0.9105
19      0.7556
20      2.2307
21      1.6372
22     -0.7227
23     -0.8980
24      1.0123
25     -0.9611
26
27  rint = 22×2
28     -2.9655    3.7684
29     -4.2697    1.9405
30     -4.1531    2.3322
31     -2.5800    4.0912
32     -0.8057    5.2671
33     -0.8182    4.0925
34     -2.3449    0.8994
35     -4.0167    2.2207
36     -1.3245    3.3492
37     -4.1449    2.2228
38
39  s = 1×4
40      0.6521    7.9661    0.0008    2.5974
41

```

$$R^2 = 0.6521$$

3.

(1). Use the forward-difference formulas and back ward-difference formulas to determine each missing entry in the following table.

| $x$ | $f(x)$   | $f'(x)$         |
|-----|----------|-----------------|
| 0.0 | 0.000 00 | 3.707           |
| 0.2 | 0.741 40 | 3.152           |
| 0.4 | 1.371 8  | 3.152(backward) |

**(2). Data in part(1) were taken from the function  $y = e^x - 2x^2 + 3x - 1$ . Compute the actual errors in part(1) , and find error bounds using the formulas.**

$$y' = e^x - 4x + 3, y'' = e^x - 4$$

$$\text{because } f'(x_0) = \frac{f(x_0+h)-f(x_0)}{h} - \frac{h}{2} f''(c) \quad c \in [x_0, x_0 + h]$$

So when the  $x_0 = 0$ , we use the forward-difference ,  $c \in [0, 0.2]$ ,  $h = 0.2$

$$\text{the Error} = \left| \frac{h}{2} f''(c) \right| \in [0.2779, 0.3]$$

when the  $x_1 = 0.2$ ,  $c \in [0.2, 0.4]$

$$\text{the Error} \in [0.2508, 0.2779]$$

the  $x_2 = 0.4$  is the same Error.